MILITARY SPECIFICATION

NOZZLES, ABRASIVE BLAST, VENTURI TYPE

1. SCOPE

1.1 Scope. - This specification covers abrasive blast nozzles of the venturi type.

1.2 Classification. - The abrasive blast nozzles shall be of the following types, as specified (see 6.1):

Type I - Heavy duty.
Type II - Light duty.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards of the issue in effect on date of invitation for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Boxes, Set-Up, Paperboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive, Paper, Water Resistant.

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MIL-P-116	Preservation, Methods of.
MIL-C-3955	Cans, Fiber, Spirally Wound.
MIL-B-4229	Boxes; Paperboard, Metal-Stayed.
MIL-B-10377	Boxes: Wood-Cleated, Veneer, Paper Overlaid.

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MIL-L-10547	Liners, Case, Waterproof.

STANDARDS

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MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, N. Y.)

NATIONAL BUREAU OF STANDARDS
Handbook H28 Screw-Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material. -

3.1.1 Barrels. - The barrels of the nozzles shall be made of corrosion-resistant treated steel or zinc base alloy.

3.1.2 Liners. - The nozzles shall be lined with not less than 5/32 inch thickness of tungsten or

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boron carbide, as specified (see 6.1). The liners shall be essentially two truncated cones joined together by a straight section. The lengths of the cones and straight sections shall be ground to provide a perfectly flat and smooth connection.

3.2 Threads. - All screw threads shall be smooth finished, without imperfections such as burrs or projections. Screw threads shall be right hand, class 2 fit, in accordance with Handbook H28, 1-1/4 inch straight pipe threads, 11-1/2 threads per inch.

3.3 Marking. - Each nozzle barrel shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined.

3.4 Venturi type. - The nozzles shall be of the venturi type, that is, the type in which the passage through the nozzle has a narrowing or constricted area utilizing the principle that the velocity of the abrasive flowing through is increased and its pressure against the nozzle liner is decreased in the narrowing of the constricted area.

3.5 Type I, heavy duty. - Type I, heavy duty nozzles shall be suitable in all respects for heavy duty cleaning where the blaster can operate the nozzle at 12 to 20 inches from the surface being blasted. The nozzles shall be of one of the sizes shown in table I, as specified (see 6.1)

Table I - Sizes for type I, heavy duty

Length	Inside diameter orifice	Inside diameter inlet	Inside diameter outlet
Inches	Inch	Inch	Inch
7-3/4	7/16	1	5/8
6-1/2	3/8	1	9/16

3.6 Type II, light duty. - Type II, light duty nozzles shall be suitable in all respects for light duty cleaning where the blaster can operate with the nozzle at 12 inches or less from the surface being blasted. The nozzles shall be of the size shown in table II.

Table II - Size for type II, light duty

Length	Inside diameter orifice	Inside diameter inlet	Inside diameter outlet
Inches	Inch	Inch	Inch
4	3/16	1	5/16

3.7 Workmanship. - All surfaces of the nozzle barrels shall be smooth and free from burrs or other harmful extraneous imperfections. The workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling for acceptance inspection. -

4.2.1 Lot. - For the purpose of acceptance inspection, a lot shall consist of not more than 500 nozzles of the same type and size offered for delivery at one time.

4.2.2 Sampling for examination. - A random sample of nozzles shall be selected from each lot in accordance with Standard MIL-STD-105 and as shown in table III. The acceptable quality level equals approximately 1.5 percent defective. Examination shall be conducted as specified in 4.3.

Table III - Sampling for visual and dimensional examination

Number of nozzles in lot	Number of nozzles in sample	Acceptance number (defectives)	Rejection number (defectives)
Under 25	10	0	1
26 to 40	15	0	1
41 to 110	25	1	2
111 to 300	35	1	2
301 to 500	50	2	3

4.3 Visual and dimensional examination. - Each of the sample nozzles selected in accordance with 4.2.2 shall be visually and dimensionally examined to determine conformance with all the requirements of this specification. Any nozzles in the sample containing one or more visual or dimensional defects shall be rejected, and if the number of defective nozzles in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

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5.1 Preservation and packaging (see 6.1). -

5.1.1 Level A. - Nozzles shall be individually preserved and packaged in accordance with method III of Specification MIL-P-116. Contact preservative is not required.

5.1.1.1 Unit containers. - Unit containers shall conform to Specification PPP-B-566, PPP-B-676, MIL-B-4229, PPP-B-636 or MIL-C-3955 at the option of the contractor, provided that the various requirements of the container specification are complied with. Sealing and closure of unit containers shall conform to the applicable container specification and appendix thereto.

5.1.1.2 Intermediate containers. - Unit quantities in an intermediate container shall be as specified (see 6.1). Intermediate containers shall conform to Specifications PPP-B-566, PPP-B-676, MIL-B-4229 or PPP-B-636 at the option of the contractor. Box closures shall conform to the applicable box specification or the appendix thereto. The gross weight of the paperboard boxes shall not exceed 10 pounds and fiber boxes shall not exceed 20 pounds.

5.1.2 Level C. - Nozzles shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2 Packing (see 6.1). -

5.2.1 Level A. - Nozzles, packaged as specified in 5.1, shall be packed in overseas type wood-cleated fiberboard, nailed wood, wirebound wood, fiber, wood-cleated veneer paper overlaid, or wood-cleated plywood boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-636 (class 3), MIL-B-10377, or PPP-B-601, respectively at the option of the contractor. Shipping containers shall have caseliners conforming to Specification MIL-L-10547 and shall be closed and sealed in accordance with the appendix thereto. Caseliners for boxes conforming to Specification PPP-B-636 may be omitted provided all joints and corners of the boxes are sealed with minimum 1-1/2 inch wide tape conforming to Specification PPP-T-76. Boxes shall be closed and strapped in accordance with the applicable box specification or appendix thereto. The gross weight of wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.2 Level B. - Nozzles, packaged as specified in 5.1, shall be packed in domestic type wood-cleated fiberboard, nailed wood, wire-bound wood, cleated plywood or wood-cleated veneer paper overlaid boxes or fiber boxes conforming to Specification PPP-B-591, PPP-B-621, PPP-B-585, PPP-B-601, MIL-B-10377 or PPP-B-636 (class 2), respectively, at the option of the contractor. Box closures shall be as specified in the applicable box specification or appendix thereto. The gross weight of the wood or wood-cleated boxes shall not exceed 200 pounds; fiber boxes shall not exceed the weight limitations of the applicable box specification.

5.2.3 Level C. - Nozzles, packaged as specified in 5.1, shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or other regulations as applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order or herein, interior and exterior shipping containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Types required (see 1.2).
(c) Whether nozzles should be lined with tungsten or boron carbide (see 3.1.2).
(d) Size of type I nozzles (see 3.5).
(e) Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).
(f) Unit quantities in intermediate container (see 5.1.1.2).

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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Navy-Bureau of Ships
(Project 4730-N033Sh)

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